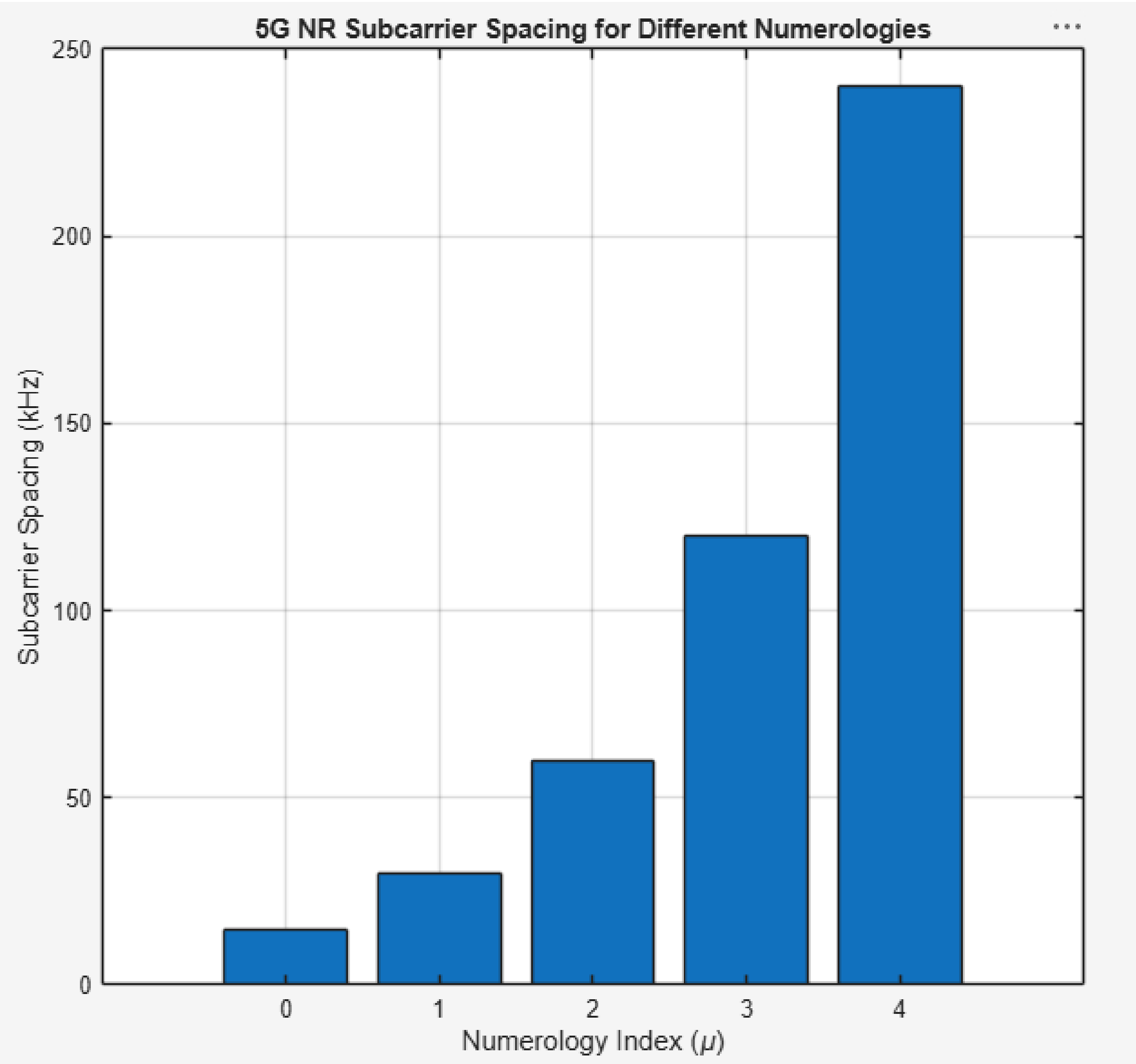
Ex.No.6: 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective 1: Matlab Code and Output

/MATLAB Drive/slotb6.m

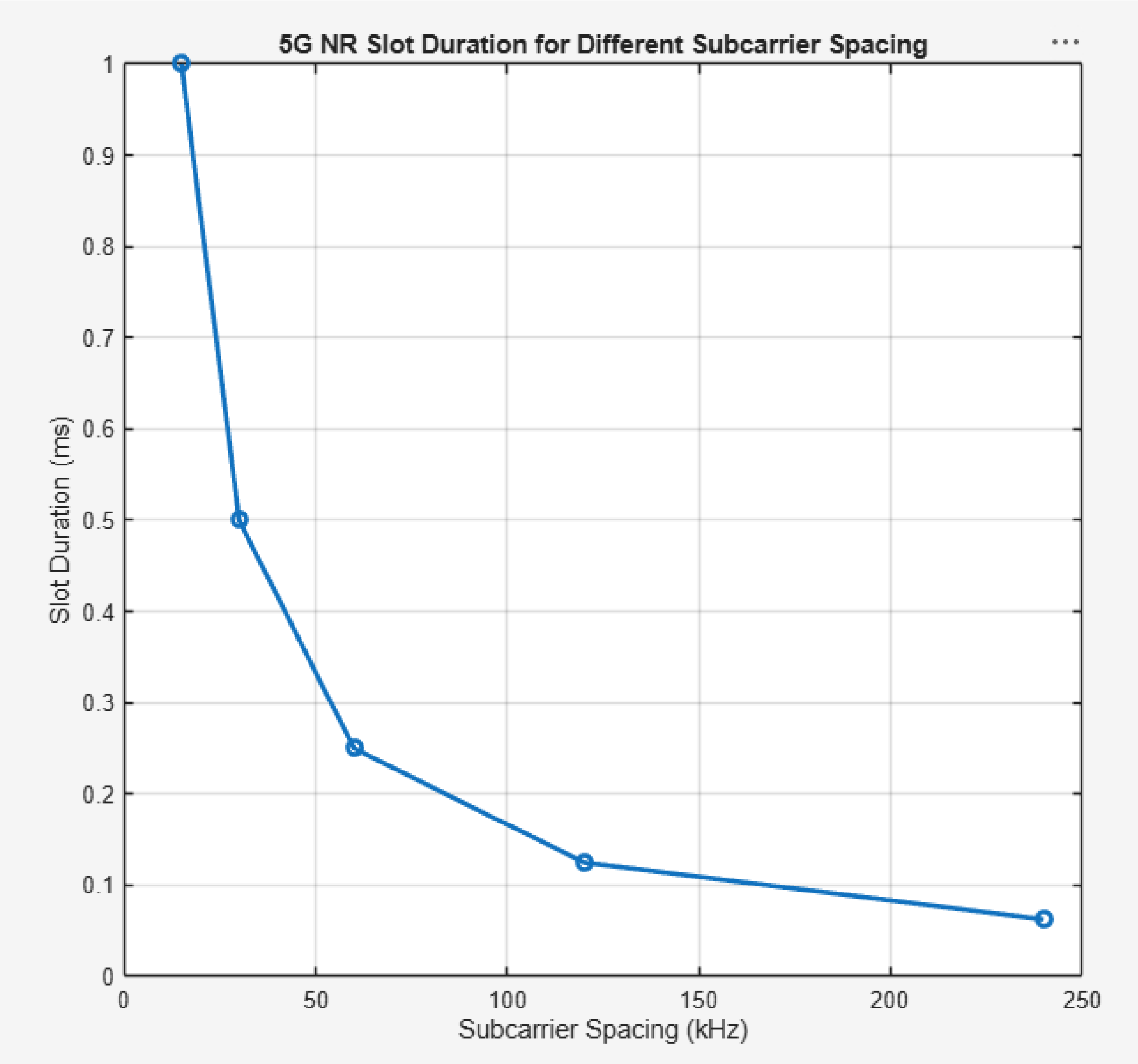
```
clc;
clear;
close all;
scs = [15 30 60 120 240];
mu = 0:4;
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
% Subcarrier Spacing (kHz)
% Numerology Index
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective 2: Matlab Code and Output

```
scs = [15 30 60 120 240];
% Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625];  % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')

```



Objective 3: Matlab Code and Output

```
scs = [15 30 60 120 240];
symbols = [14 14 14 14 14];
figure
bar(scs, symbols)
% Subcarrier Spacing (kHz)
% OFDM Symbols per Slot
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```